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| Sensor and control points for HNP Kibera Town Centre: November 2014 |                         | Diagram "Water Treatment" showing detail of A/A, Sand filters, and Regeneration Tank 600L: WT |                    |                        |                       | Includes Diagram "Waste Recycling System" : WR |                     |                                                     | Last updated: JG 23 April 2018                             |                    | Version 29 October 2014 1am |
| Equipment Count                                                     | Pumps: 13               | Pump Motor Sensors: 13                                                                        | On line Sensors: 9 | Tanks and Vaults: 11   | Tank Level Sensors: 4 | Chlorine Sensors: 2                            | Fluoride Sensors: 0 | Well sensors: 0                                     | Acid, alkalinity sensors: 2?; pH: 1; ORP: ?; TDS: 1;       | Filters: 4         |                             |
| Color Key:                                                          | Pump                    | Pipe                                                                                          | Tank               | Vault                  | Control               | Sensor                                         | Filter / other      | Possible Future Sensor                              | 1" = 25mm<br>2" = 50mm<br>4" = 100mm                       | 6" = 150mm         |                             |
|                                                                     |                         |                                                                                               |                    |                        |                       |                                                |                     |                                                     |                                                            |                    |                             |
|                                                                     | Element                 | Element ID                                                                                    | XIO ID Number      | Location               | Measuring             | Units Measured                                 | Vendor              | Characteristics                                     | Purpose                                                    | Inputs and Outputs |                             |
| Power Group                                                         |                         |                                                                                               |                    |                        |                       |                                                |                     |                                                     |                                                            |                    |                             |
| Kenya Power: Grid Connection                                        |                         |                                                                                               |                    | Power Room             | Power, Energy         | Volts, Amps, Amp-hours; VAR                    |                     |                                                     |                                                            |                    |                             |
| Kenya Power: billing meter                                          | KPL Tariff Meter        |                                                                                               |                    | Power Room             | Energy                | kWh                                            |                     | Hand read daily by KTC; monthly by KPL              | Billing amount: usually 6 week delay                       |                    |                             |
| KP:Real Power::kW                                                   |                         |                                                                                               |                    | Power Room             | Real Power            | kW                                             |                     |                                                     |                                                            |                    |                             |
| KP: Reactive Power: VAR                                             |                         |                                                                                               |                    | Power Room             | Reactive Power: Q     | VAR                                            |                     |                                                     |                                                            |                    |                             |
| KP: Power Factor                                                    |                         |                                                                                               |                    | Power Room             | Power Factor          | PF                                             |                     |                                                     |                                                            |                    |                             |
| KP: Current                                                         |                         |                                                                                               |                    | Power Room             | Current               | Amps                                           |                     |                                                     |                                                            |                    |                             |
| Diesel Generator                                                    |                         |                                                                                               |                    | Power Room             | Power, Energy         | Amps                                           |                     |                                                     |                                                            |                    |                             |
| Diesel Generator: current                                           | Diesel ouput current    |                                                                                               | D7                 | Power Room             | Energy                | kWh                                            |                     |                                                     | might be a mistake in XIO label                            | 4-20MA to XIO?     |                             |
| E24 Inverter                                                        |                         |                                                                                               |                    |                        | Power, Energy         |                                                |                     |                                                     |                                                            |                    |                             |
| E24 Battery Level                                                   |                         |                                                                                               |                    |                        | Energy                |                                                |                     |                                                     |                                                            |                    |                             |
| E24 PV Current                                                      |                         |                                                                                               |                    |                        | Power                 |                                                |                     |                                                     |                                                            |                    |                             |
|                                                                     |                         |                                                                                               |                    |                        |                       |                                                |                     |                                                     |                                                            |                    |                             |
| Potable Water Group                                                 |                         |                                                                                               |                    |                        |                       |                                                |                     |                                                     |                                                            |                    |                             |
| Borehole Group                                                      |                         |                                                                                               |                    |                        |                       |                                                |                     |                                                     |                                                            |                    |                             |
| Borehole Pump                                                       | Pump                    | P BH1                                                                                         |                    | Base of borehole       |                       |                                                | Grundfos SP17-29    | 18.5kW; 19M3/hr; 206m head; Power line: 3-wire feed | Push accumulated bore hole water to surface                |                    |                             |
| Borehole Pump Current                                               | Pump Motor Sensor:Amps  | S BH Current                                                                                  | D5                 | Power Room Breaker Box | Amps                  | Amps                                           |                     |                                                     |                                                            |                    |                             |
|                                                                     | Pump Control            | PC BH1C1                                                                                      |                    | At pump                |                       |                                                |                     | Pump Control                                        | Control speed of pump                                      |                    |                             |
|                                                                     | Pump Motor Sensor: Temp | S BH1S1                                                                                       |                    | Base of borehole       | motor temperature     | C°                                             |                     |                                                     | Monitor motor performance                                  |                    |                             |
|                                                                     | Sensor: turbidity       | S BH1S3                                                                                       |                    | Base of borehole       | Turbidity             |                                                |                     | Optical sensor                                      | Detect changes in bore hole turbidity coming from aquifers |                    |                             |
|                                                                     | Sensor: Water Temp      | S BH1S4                                                                                       |                    | Base of borehole       | Water Temperature     | C°                                             |                     | Temp sensor                                         | Track water temp                                           |                    |                             |

|                                              |                           |               |                     |                                |                                             |                                                               |  |                                                                                  |                                                                                                                                                          |                                |
|----------------------------------------------|---------------------------|---------------|---------------------|--------------------------------|---------------------------------------------|---------------------------------------------------------------|--|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| Well Water Level                             | Sensor: well water level  | S BH1S5       |                     | Water Level                    | Depth from Well Head                        | cm                                                            |  | Continual depth Measurement:<br>Pressure sensor?<br>Acoustic sensor?<br>Bubbler? | Calculate recharge rates of feed aquifers; use to track impact of other boreholes near us                                                                |                                |
| Aquifer intake 1                             | Shaft                     | A BH1A1       |                     | Aquifer slots in well casing   | Flow rate                                   | L / M                                                         |  | Intake at 160 M;<br>Pressure sensor                                              | Measure Aquifer 1 flow                                                                                                                                   |                                |
| Aquifer intake 2                             | Shaft                     | A BH1A2       |                     | Aquifer slots                  | Flow rate                                   | L / M                                                         |  | Intake at 210M;<br>Pressure sensor                                               | Measure Aquifer 2 flow                                                                                                                                   |                                |
|                                              |                           |               |                     |                                |                                             |                                                               |  |                                                                                  |                                                                                                                                                          |                                |
|                                              | Top of Shaft              |               |                     | Well Head                      |                                             |                                                               |  | Pressure sensor<br>Mag flow?                                                     | Flow rate from borehole; detect leaks                                                                                                                    |                                |
|                                              |                           |               |                     |                                |                                             |                                                               |  |                                                                                  |                                                                                                                                                          |                                |
| Pipe from Well Head to Borehole Storage Tank | Pipe                      | Pipe 1A       |                     | From Well Head to Ground Floor |                                             |                                                               |  | 2" pipe?; 50mm?                                                                  |                                                                                                                                                          |                                |
| WERMA Flow Meter                             | Borehole Flow Sensor      | S BH-Flow1    | Visually Read Daily | Well Head                      | Flow rate                                   | GPM<br>L / M                                                  |  | Rotary sensor                                                                    | Calculate well pump flow rate: (here, or at top of well)                                                                                                 |                                |
| Borehole Flow Meter                          | Borehole Flow Sensor      | S BH-Flow2    | D16                 | Base of T1                     | Flow rate                                   | GPM<br>L / M                                                  |  | Rotary sensor                                                                    | Calculate well pump flow rate: (here, or at top of well)                                                                                                 |                                |
| Borehole Storage Tank                        | Tank                      | BST 1         |                     | Ground Floor                   |                                             |                                                               |  | 5000 L; dimensions: replaced once for cracks;                                    |                                                                                                                                                          | Pipes in: 1;<br>Pipes out: 1   |
| Borehole Tank Level                          | Pressure Sensor           | S-BST Level 1 | D10                 | Ground Floor                   | weight of water => depth of Borehole Tank 1 | cm                                                            |  |                                                                                  |                                                                                                                                                          | 4-20 MA loop to XIO box 1?     |
| Borehole Tank Level                          | Float Valve               | S-BST Level 2 |                     | Ground Floor                   | Borehole Tank 1 Level                       | water level compared to set point                             |  |                                                                                  |                                                                                                                                                          | 4-20 MA loop to Borehole Pump? |
| Pipe to Potable Supply Treatment Group       | Pipe from T1 to Treatment | Pipe 1B       |                     |                                |                                             |                                                               |  |                                                                                  |                                                                                                                                                          |                                |
|                                              |                           |               |                     |                                |                                             |                                                               |  |                                                                                  |                                                                                                                                                          |                                |
| Potable Treatment Group                      |                           |               |                     |                                |                                             | Needs cleanup, There are duplicate units with different names |  |                                                                                  | Bring fluoride level below 0.8ppm; balance pH; soften; chlorinate; check for pathogen intrusion; do full ionic, biological, chemical, and pollutant test |                                |

|                                   |                   |          |     |                             |                               |          |           |                                                                            |                                                                                                                                                                                        |    |
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| Pipe                              | Pipe              | Pipe 2A  |     | Ground Floor Treatment Room |                               |          |           | Incoming from T 1; manifold to two pumps: Duty Standby CR 15-5             |                                                                                                                                                                                        |    |
| Duty Standby CR 15-5: Pump 1      | Pump              | P 2B     |     | Ground Floor Treatment Room |                               |          | Grundfos  |                                                                            | Measure flow...Need to measure ways to initiate Backwash Cycle                                                                                                                         | WT |
|                                   | Pump Control      | C P2B C1 |     | Ground Floor Treatment Room | Controls                      |          |           |                                                                            | Control pump speed                                                                                                                                                                     |    |
|                                   | Pump Motor Sensor | S P2B S1 | D8  | Ground Floor Treatment Room | Power draw                    | Amps     |           | One of these pumps has an inductive sensor on its power feed               | Monitor motor performance                                                                                                                                                              |    |
| Duty Standby CR 15-5: Pump 2      | Pump              | P 2C     |     | Ground Floor Treatment Room |                               |          | Grundfos  |                                                                            | Measure flow...Need to measure ways to initiate Backwash Cycle                                                                                                                         | WT |
|                                   | Pump Control      | C P2C C1 |     | Ground Floor Treatment Room | Controls                      |          |           |                                                                            | Control pump speed                                                                                                                                                                     |    |
|                                   | Pump Motor Sensor | S P2C S1 | D8  | Ground Floor Treatment Room | Power draw                    | Amps     |           | One of these pumps has an inductive sensor on its power feed               | Monitor motor performance                                                                                                                                                              |    |
| Dosing Tank: T 170L Chlorine?     | Tank              | T 2A     |     | Ground Floor Treatment Room |                               |          |           | Chlorine inserted before Pumps and Sand Filters: 170L                      | Chlorine? kill biofilm? Is this really before the sand filters? Before Activated Alumina? What is relationship to Base or Alkiline insertion to keep pH correct for Activated Alumina? |    |
| Hypochlorite Tank Level           | Tank Depth Sensor | T 2AS1   | D12 | Ground Floor Treatment Room | Tank level                    | mm       | ?         | Ultrasonic sensor?                                                         | Measure level of Hypochlorite Solution                                                                                                                                                 |    |
| Dosing Tank Pump: T 170L Chlorine | Pump              | P 2A     |     | Ground Floor Treatment Room |                               |          |           | Pumps Hypochlorite mixture into flow. Does pump exist? or gravity-fed?     | Control Chlorine flow                                                                                                                                                                  |    |
|                                   | Pump Control      | C P2AC1  |     | Ground Floor Treatment Room |                               |          |           |                                                                            | Control pump speed                                                                                                                                                                     | WT |
|                                   | Pump Motor Sensor | S P2AS1  |     | Ground Floor Treatment Room | Power draw, motor temperature | Amps, C° |           |                                                                            | Monitor motor performance                                                                                                                                                              | WT |
| Sensors marked 3E: Chlorine level | Sensor            | S T2AS1  |     | Ground Floor Treatment Room | Chlorine level                | ppm      | ATI: Q45C | either photometrically or amperimetric free chlorine sensor (uses reagent) | Measure ppm of Chlorine                                                                                                                                                                | WT |

|                                           |                   |         |  |                             |                               |         |  |                                                            |                                                                                |    |
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| Pipe to manifold to both sand filters     | Pipe              | Pipe 2B |  | Ground Floor Treatment Room |                               |         |  |                                                            |                                                                                |    |
| Pipe manifold to both sand filters        | Pipe              | Pipe 2C |  | Ground Floor Treatment Room |                               |         |  | Each end terminates in a valve box before each Sand Filter |                                                                                |    |
| Pipe into Sand Filter 1                   |                   | Pipe 2D |  | Ground Floor Treatment Room |                               |         |  |                                                            |                                                                                |    |
| Sand Filter 1: CXD 900                    | Sand Filter 1     | F 2A    |  | Ground Floor Treatment Room | ?                             |         |  |                                                            | Pipes in: 1;<br>Pipes out: 1;<br>Remove colloids, any incursion; build biofilm | WT |
| Pipe out of Sand Filter 1                 | Pipe              | Pipe 2E |  | Ground Floor Treatment Room |                               |         |  |                                                            |                                                                                |    |
| Pipe into Sand Filter 2                   | Pipe              |         |  | Ground Floor Treatment Room |                               |         |  |                                                            |                                                                                |    |
| Sand Filter 2: CXD 900                    | Sand Filter 2     | F 2B    |  | Ground Floor Treatment Room | ?                             |         |  |                                                            | Pipes in: 1;<br>Pipes out: 1;<br>Remove colloids, any incursion; biofilm       | WT |
| Pipe out of Sand Filter 2                 |                   | Pipe 2F |  | Ground Floor Treatment Room |                               |         |  |                                                            |                                                                                |    |
| Sensor marked 3A after Sand Filters       |                   | S 2B    |  | after sand filter           | Flow rate                     | L / M   |  |                                                            | Flow where pipes join from sand filters                                        |    |
| Pipe from Sand Filters to Alumina Filters |                   |         |  |                             |                               |         |  |                                                            |                                                                                |    |
| T170 L Acid input                         | Tank              | T 2D    |  | Ground Floor Treatment Room | Ph level                      |         |  |                                                            | Acid input for proper pH                                                       | WT |
|                                           | Tank Sensor       | S T2DS1 |  |                             | Tank Level                    | mm      |  | Ultrasonic sensor                                          | Tank Level                                                                     |    |
| Sensor marked 3D                          | Sensor            | S 2C    |  | between T170L Acid input    | Flow of acid control?         |         |  |                                                            | ?                                                                              |    |
| Pipe from Acid Input to flow              | Pipe              | Pipe 2G |  | Ground Floor Treatment Room |                               |         |  |                                                            |                                                                                |    |
| Pipe from flow to Davey Pump              | Pipe              | Pipe 2H |  | Ground Floor Treatment Room |                               |         |  |                                                            |                                                                                |    |
| GRUNDFOS CM 10-9                          | Pump              | P 2D    |  | Ground Floor Treatment Room |                               |         |  | Does this pressurize line up to High Potable Tank?         | Pumps in and out of Regeneration Tank                                          | WT |
|                                           | Pump Control      | C P2DC1 |  | Ground Floor Treatment Room | Controls                      |         |  |                                                            |                                                                                |    |
|                                           | Pump Motor Sensor | S P2DS1 |  | Ground Floor Treatment Room | Power draw, motor temperature | Amps,C° |  |                                                            | Monitor motor performance                                                      |    |

|                                                    |                    |         |  |                             |                               |         |                                         |                                                                                                   |                                                                 |                           |
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| Pipe from Davey Pump to Regeneration Tank          |                    | Pipe 2I |  | Ground Floor Treatment Room |                               |         |                                         |                                                                                                   |                                                                 |                           |
| Regeneration Tank T600L after A/A filters          | Tank               | T 2E    |  | Ground Floor Treatment Room |                               |         |                                         |                                                                                                   | 600L tank; where mixing of acid and base softeners takes place? | Pipes in: 1; Pipes out: 1 |
|                                                    | Tank Sensor        | S T2ES1 |  |                             | Tank Level                    | mm      |                                         | Ultrasonic sensor                                                                                 | Level of Regeneration Tank                                      |                           |
| Pipe from Regeneration Tank to A/A Filter Manifold |                    | Pipe 2J |  |                             |                               |         |                                         |                                                                                                   |                                                                 |                           |
| T170 L Base input                                  | Tank               | T2F     |  | Ground Floor Treatment Room |                               |         |                                         |                                                                                                   | Base input for proper pH                                        | Pipes in: 1; Pipes out: 1 |
|                                                    | Tank Sensor        | S T2FC1 |  |                             | Tank Level                    | mm      |                                         | ultrasonic sensor                                                                                 |                                                                 |                           |
| Sensor marked 3C                                   | Sensor             | S 2D    |  |                             | pH level                      | pH      | ATI: Analytical Technologies, Inc: Q45P |                                                                                                   | Measure pH                                                      | WT                        |
| Pipe: A/A Filter Manifold                          |                    | Pipe 2K |  | Ground Floor Treatment Room |                               |         |                                         |                                                                                                   |                                                                 |                           |
| Activated Alumina Filter 1                         | Filter             | F2A     |  | Ground Floor Treatment Room |                               |         |                                         |                                                                                                   | Pipes in: 1; Pipes out: 1 used also for backwash                | WT                        |
| Activated Alumina Filter 2                         | Filter             | F2B     |  | Ground Floor Treatment Room |                               |         |                                         |                                                                                                   | Pipes in: 1; Pipes out: 1 used also for backwash                | WT                        |
| Sensor 3B: Fluoride level?                         | Manual Measurement | S2E     |  | Ground Floor Treatment Room | Fluoride level                | ppm     | ?                                       | Manual use of fluoride hand-held meter. Amperimetric? Does it use any reagents? Can it be online? | Measure fluoride after Fluoride reduction                       | WT                        |
| Backwash pipe into A/A filters                     | Pipe               | Pipe 2L |  | Ground Floor Treatment Room |                               |         |                                         |                                                                                                   |                                                                 | WT                        |
| REGENERATION PUMP GRUNDFOS CM10-9                  | Pump               | P2E     |  | Ground Floor Treatment Room |                               |         |                                         | ?                                                                                                 | How is this controlled? Who sets backwash schedule?             |                           |
|                                                    | Pump Control       | P2EC1   |  | Ground Floor Treatment Room |                               |         |                                         |                                                                                                   | Control Backwash pump                                           |                           |
|                                                    | Pump Motor Sensor  | P2ES1   |  | Ground Floor Treatment Room | Power draw, motor temperature | Amps,C° |                                         |                                                                                                   | Monitor motor performance                                       |                           |
| Pipe out: Treated Water                            | Pipe               | Pipe 2M |  | Ground Floor Treatment Room |                               |         |                                         | to High Potable Tank                                                                              |                                                                 |                           |

|                                                                                                    |                    |                  |            |                                |                                                          |       |  |                                                             |                                                                            |                                  |
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| Pipe out:<br>Backwash to<br>Sewer Line                                                             | Pipe               | Pipe 2N          |            | Ground Floor<br>Treatment Room |                                                          |       |  | to Sewer Line or to<br>Backwash Vault                       |                                                                            |                                  |
| New testing<br>equipment,<br>tanks for<br>testing<br>alternative<br>bauxite<br>treatment<br>system |                    |                  |            | Ground Floor<br>Treatment Room |                                                          |       |  |                                                             |                                                                            |                                  |
|                                                                                                    |                    |                  |            |                                |                                                          |       |  |                                                             |                                                                            |                                  |
|                                                                                                    |                    |                  |            |                                |                                                          |       |  |                                                             |                                                                            |                                  |
|                                                                                                    |                    |                  |            |                                |                                                          |       |  |                                                             |                                                                            |                                  |
|                                                                                                    |                    |                  |            |                                |                                                          |       |  |                                                             |                                                                            |                                  |
| <b>Potable Supply Distribution Group</b>                                                           |                    |                  |            |                                |                                                          |       |  |                                                             |                                                                            |                                  |
| 0                                                                                                  | Pump               | P 3A             |            | Ground Floor<br>Treatment Room |                                                          |       |  | 3-Phase: is this the<br>same Davey pump P<br>2D?            | Move treated potable<br>water to first elevated<br>storage tank            |                                  |
|                                                                                                    | Pump Control       | C P3AC1          |            | Ground Floor<br>Treatment Room |                                                          |       |  |                                                             | Control speed of pump<br>coming from float sensor<br>in High Clean Tank 1? |                                  |
|                                                                                                    | Pump Sensor        | S P3AS1          |            | Ground Floor<br>Treatment Room | Temp,<br>current                                         | C°, A |  |                                                             | Operating parameters of<br>pump                                            |                                  |
| High Clean<br>Tank 1                                                                               | Tank               | T3A              |            | First Floor West               |                                                          |       |  | 5000L; use tank float<br>switch                             | Store treated water                                                        | Pipes in: 1<br>Pipes out: 2      |
| Not Installed                                                                                      | Tank Sensor        | T3AS1            |            | First Floor West               | Tank Level                                               | mm    |  | Ultrasonic sensor                                           | Tank Level                                                                 |                                  |
| High Clean<br>Tank 1 Level                                                                         | Pressure<br>Sensor | S-HCT Level<br>1 | <b>D10</b> | First Floor West               | weight of<br>water ==><br>depth of<br>Borehole<br>Tank 1 | cm    |  | Is the tank level<br>measured in High<br>Tank, or Low Tank? |                                                                            | 4-20 MA loop<br>to XIO box<br>1? |
| Pipe from High<br>Clean Tank 1 to<br>Low Clean<br>Tank 2                                           | Pipe               | Pipe T1T2        |            | First Floor to<br>Ground Floor |                                                          |       |  | Gravity feed to lower<br>Potable Tank                       |                                                                            |                                  |
| Pipe from High<br>Clean Tank 1 to<br>Hot Water<br>Tanks                                            | Pipe               | Pipe T1Hot       |            | First Floor to<br>First Floor  |                                                          |       |  | Gravity feed                                                | to hotwater pump??                                                         |                                  |
| Low Clean<br>Tank 2                                                                                | Tank               | T3B              |            | Ground Floor                   |                                                          |       |  | 5000L; use tank float<br>switch                             |                                                                            | Pipes in: 1<br>Pipes out: 1      |
| Low Clean<br>Tank 2 Level                                                                          | Pressure<br>Sensor | S-HCT Level<br>1 | <b>D10</b> | Ground Floor                   | weight of<br>water ==><br>depth of<br>Borehole<br>Tank 1 | cm    |  | Is the tank level<br>measured in High<br>Tank, or Low Tank? |                                                                            | 4-20 MA loop<br>to XIO box<br>1? |



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| <b>Potable Water Heating, Temp Measurement, Mixing and Control Group</b> |                       |           |    |                     |               |           |  |                                                                                 | Final measurement of water quality; Temperature control for various destinations |                           |
| Pipe from Potable Tank 1 to Hot Water Manifold input to Hot Water Tanks  |                       |           |    | First Floor         |               |           |  | Cold water input will be heated by closed loop solar system or electric heaters |                                                                                  |                           |
| Hot Water Flow Meter                                                     | Hot Water Flow Sensor | S HW-Flow | D2 | First Floor         | Flow rate     | GPM L / M |  | Rotary sensor                                                                   | Calculate hot water flow rate: going to Men's, Women's. Laundry?                 |                           |
| Solar Hot Water Collector Bank 1                                         | Tank                  |           |    |                     |               |           |  | 10 collectors for each tank; pump circulates closed-loop liquid                 |                                                                                  | Pipes out: 1; pipes in: 1 |
| Solar Hot Water Collector Bank 2                                         | Tank                  |           |    |                     |               |           |  | 10 collectors for each tank; pump circulates closed-loop liquid                 |                                                                                  | Pipes out: 1; pipes in: 1 |
| Solar Hot Water Collector Bank 3                                         | Tank                  |           |    |                     |               |           |  | 10 collectors for each tank; pump circulates closed-loop liquid                 |                                                                                  | Pipes out: 1; pipes in: 1 |
| Solar Hot Water Collector Bank 4                                         | Tank                  |           |    |                     |               |           |  | 10 collectors for each tank; pump circulates closed-loop liquid                 |                                                                                  | Pipes out: 1; pipes in: 1 |
|                                                                          |                       |           |    |                     |               |           |  |                                                                                 |                                                                                  |                           |
| Pipes from roof to hot water tanks                                       | Pipe                  |           |    | Roof to First Floor |               |           |  | Closed loop system of heat transfer liquid to heat exchangers in each tank      |                                                                                  |                           |
| H2O Solar                                                                | Sensor                | S4C       |    |                     | Temp          | C°        |  |                                                                                 | Monitor rooftop solar heating units                                              |                           |
| H2O Solar pump GRUNDFOS CM3-4                                            | Pump                  | P4A       |    | Roof                |               |           |  |                                                                                 | Move solar thermal hot water                                                     |                           |
| H2O Solar pump control                                                   | Pump Control          | P4AC1     |    | Roof                |               |           |  |                                                                                 | Control speed of pump                                                            |                           |
|                                                                          | Pump Sensor           | P4AS1     |    | Roof                | Temp, current | C°, A     |  |                                                                                 | Operating parameters of pump                                                     |                           |
| H2O Heater                                                               |                       |           |    | First Floor         |               |           |  |                                                                                 | External heating unit (electric)                                                 |                           |
| H2O Heater Control                                                       | Control               | C4A       |    | First Floor         |               |           |  |                                                                                 | Turn on external heater                                                          |                           |
| H2O Heater pump                                                          | Pump                  | P4B       |    | First Floor         |               |           |  |                                                                                 |                                                                                  |                           |

|                                                         |              |       |  |             |                  |               |  |                                                   |                                 |                             |
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|                                                         | Pump Control | P4BC1 |  | First Floor |                  |               |  |                                                   | Control speed of pump           |                             |
|                                                         | Pump Sensor  | P4BS1 |  | First Floor | Temp,<br>current | C°, A         |  |                                                   | Operating parameters of<br>pump |                             |
| Hot Water Tank 1                                        | Tank         |       |  | First Floor |                  |               |  | 1000L?                                            |                                 | Pipes in: 1<br>Pipes out: 1 |
| Pipe to Hot Water Manifold out                          | Pipe         |       |  | First Floor |                  |               |  | 50mm?                                             |                                 |                             |
| Hot Water Tank                                          | Tank         |       |  | First Floor |                  |               |  | 1000L?                                            |                                 | Pipes in: 1<br>Pipes out: 1 |
| Pipe to Hot Water Manifold out                          | Pipe         |       |  | First Floor |                  |               |  | 50mm?                                             |                                 |                             |
| Hot Water Tank                                          | Tank         |       |  | First Floor |                  |               |  | 1000L?                                            |                                 | Pipes in: 1<br>Pipes out: 1 |
| Pipe to Hot Water Manifold out                          | Pipe         |       |  | First Floor |                  |               |  | 50mm?                                             |                                 |                             |
| Hot Water Tank                                          | Tank         |       |  | First Floor |                  |               |  | 1000L?                                            |                                 | Pipes in: 1<br>Pipes out: 1 |
| Pipe to Hot Water Manifold out                          | Pipe         |       |  | First Floor |                  |               |  | 50mm?                                             |                                 |                             |
| Hot Water Tank                                          | Tank         |       |  | First Floor |                  |               |  | 1000L?                                            |                                 | Pipes in: 1<br>Pipes out: 1 |
| Pipe to Hot Water Manifold out                          | Pipe         |       |  | First Floor |                  |               |  |                                                   |                                 |                             |
| Pipe: Hot Water Manifold out                            | Pipe         |       |  | First Floor |                  |               |  |                                                   |                                 |                             |
| Hot Water Temperature going to all showers, sinks, cafe | Sensor       |       |  | First Floor | Temperature      | C°            |  | Really need to see if hot water system is working |                                 |                             |
| Pump Hot Water 1                                        | P5           |       |  | First Floor |                  |               |  |                                                   |                                 |                             |
|                                                         | Pump Control | P5C1  |  | First Floor |                  | Speed, On/off |  |                                                   | Control speed of pump           |                             |
|                                                         | Pump Sensor  | P3S1  |  | First Floor | Temp,<br>current | C°, A         |  |                                                   | Operating parameters of<br>pump |                             |
| Pump Hot Water 2                                        | P6           |       |  | First Floor |                  |               |  |                                                   |                                 |                             |
|                                                         | Pump Control | P5C1  |  | First Floor |                  | Speed, On/off |  |                                                   | Control speed of pump           |                             |
|                                                         | Pump Sensor  | P5S1  |  | First Floor | Temp,<br>current | C°, A         |  |                                                   | Operating parameters of<br>pump |                             |
| Pipes to distribute Hot Water                           | Pipe         |       |  | First Floor |                  |               |  |                                                   |                                 |                             |
|                                                         |              |       |  |             |                  |               |  |                                                   |                                 |                             |
| Potable Water Outside Distribution Group                |              |       |  |             |                  |               |  |                                                   |                                 |                             |

[illegible]

